

Chapter 4: Vector Spaces

MATH 3260 Linear Algebra

Sections 4.1–4.8

Chapter goals

Develop the algebra and geometry of vectors and abstract vector spaces; analyze subspaces, spanning, independence, bases, dimension, rank, nullity, coordinates, and change of basis; and apply these ideas to differential equations and conic rotations.

Original-example policy

All worked examples and numerical values are original. Themes involving the Spurs, hiking, badminton, travel, driving, Backstreet Boys, and R&B music are used selectively. Fictional data are teaching models.

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1 Vectors in $\mathbb{R}^n$

Learning objectives

Represent vectors geometrically; perform vector operations; solve vector equations; and write vectors as linear combinations.

Vector operations

For $\mathbf{u} = (u_1, \dots, u_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$,

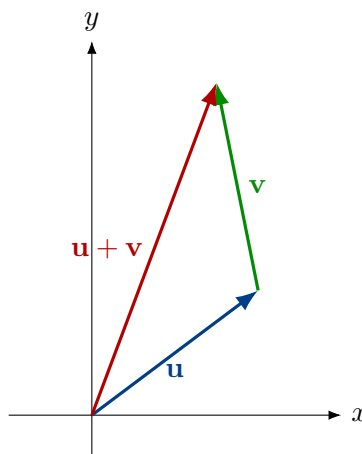
$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \dots, u_n + v_n), \quad c\mathbf{u} = (cu_1, \dots, cu_n).$$

Hiking displacement

For $\mathbf{u} = (4, 3)$ and $\mathbf{v} = (-1, 5)$,

$$\mathbf{u} + \mathbf{v} = (3, 8).$$

The sum is the net displacement.



Vector equation in $\mathbb{R}^4$

Let $\mathbf{u} = (2, -1, 3, 0)$, $\mathbf{v} = (1, 4, -2, 5)$, and $\mathbf{w} = (-1, 2, 1, -3)$. From

$$3(\mathbf{x} + \mathbf{w}) = 2\mathbf{u} - \mathbf{v} + \mathbf{x}$$

we get

$$2\mathbf{x} = 2\mathbf{u} - \mathbf{v} - 3\mathbf{w}, \quad \mathbf{x} = \left(3, -6, \frac{5}{2}, 2\right).$$

Detailed linear combination

For

$$\mathbf{v}_1 = (1, 2, 0), \quad \mathbf{v}_2 = (0, 1, 1), \quad \mathbf{v}_3 = (2, 0, 1),$$

solve $\mathbf{x} = (7, 4, 5) = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$. The augmented matrix reduces as

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 7 \\ 2 & 1 & 0 & 4 \\ 0 & 1 & 1 & 5 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 7 \\ 0 & 1 & -4 & -10 \\ 0 & 1 & 1 & 5 \end{array} \right] \xrightarrow{R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 7 \\ 0 & 1 & -4 & -10 \\ 0 & 0 & 5 & 15 \end{array} \right].$$

Thus $(c_1, c_2, c_3) = (1, 2, 3)$.

Closing note

Vector operations are the computational foundation for spans, bases, coordinates, and rank.

1.1 Definitions and terminology

Term	Meaning
Vector	Ordered list of components or directed displacement.
Component	One coordinate of a vector.
Scalar multiple	Every component multiplied by the same scalar.
Linear combination	Sum $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$.

2 Vector Spaces

Learning objectives

State the vector-space axioms, recognize standard examples, and disprove vector-space status efficiently.

Vector space

A set V with addition and scalar multiplication is a vector space if it satisfies closure, commutativity and associativity of addition, additive identity and inverses, closure under scalar multiplication, two distributive laws, compatibility, and scalar identity.

Important examples include $\mathbb{R}^n$, P_n , $M_{m,n}$, and $C[a, b]$.

P_2

The sum and scalar multiple of polynomials of degree at most two remain in P_2 . The zero polynomial and additive inverses are included, so P_2 is a vector space.

Exactly degree two is not a vector space

Both $x^2 + x$ and $-x^2 + 2$ have degree two, but their sum $x + 2$ does not. Closure under addition fails.

Closing note

Vectors may be matrices, functions, polynomials, or data objects. What matters is the linear algebra of the specified operations.

2.1 Definitions and terminology

Term	Meaning
Vector space	Set satisfying ten addition and scalar-multiplication axioms.
Closure	Operations keep results in the set.
P_n	Polynomials of degree at most n .
$M_{m,n}$	Real $m \times n$ matrices.

3 Subspaces of Vector Spaces

Learning objectives

Apply the subspace test, analyze homogeneous constraints, and distinguish linear from affine sets.

Subspace test

A nonempty $W \subseteq V$ is a subspace iff it is closed under addition and scalar multiplication. Equivalently, $a\mathbf{u} + b\mathbf{v} \in W$ whenever $\mathbf{u}, \mathbf{v} \in W$.

Plane through the origin

For $W = \{(x, y, z) : x - 2y + z = 0\}$,

$$(x, y, z) = s(2, 1, 0) + t(-1, 0, 1).$$

The homogeneous condition is preserved by linear combinations, so W is a subspace.

Affine plane

The set $x - 2y + z = 1$ is not a subspace because it does not contain the zero vector.

Intersection

The intersection of $x + y + z = 0$ and $x - z = 0$ is

$$\{t(1, -2, 1) : t \in \mathbb{R}\},$$

a line through the origin.

Closing note

Homogeneous linear conditions produce subspaces; nonzero right-hand sides generally produce affine shifts.

3.1 Definitions and terminology

Term	Meaning
Subspace	Subset that is a vector space under inherited operations.
Affine set	Shifted linear set, usually not through the origin.
Intersection	Vectors belonging to both subspaces.

4 Spanning Sets and Linear Independence

Learning objectives

Test membership in a span, determine whether sets span a space, and test linear independence.

Span

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k\}.$$

For

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix},$$

$\det(A) = 5 \neq 0$, so the columns span $\mathbb{R}^3$.

Linear independence

A set is independent if $c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0}$ has only the trivial solution.

Dependence relation

For $\mathbf{a} = (1, 2, 3)$, $\mathbf{b} = (2, 1, 0)$, and $\mathbf{c} = (5, 4, 3)$,

$$\mathbf{a} + 2\mathbf{b} - \mathbf{c} = \mathbf{0}.$$

Therefore the set is dependent.

Polynomial dependence

In P_2 , $p_3 = 1 + 2x + x^2$ equals $(1 + x) + (x + x^2)$, so the three polynomials are dependent.

Closing note

Span measures completeness; independence measures redundancy.

4.1 Definitions and terminology

Term	Meaning
Span	All linear combinations of a set.
Spanning set	Set whose span is the target space.
Linearly independent	Only trivial homogeneous coefficient solution.
Linearly dependent	Has a nontrivial dependence relation.

5 Basis and Dimension

Learning objectives

Recognize bases, extract bases from spanning sets, and compute dimension.

Basis

A basis is a linearly independent spanning set. Every vector has a unique representation relative to a basis.

The set

$$\{(1, 2, 0), (0, 1, 1), (2, 0, 1)\}$$

is a basis for $\mathbb{R}^3$ because its basis matrix has determinant 5.

Dimension

The dimension of a finite-dimensional vector space is the number of vectors in any basis.

$$\dim(\mathbb{R}^n) = n, \quad \dim(P_n) = n + 1, \quad \dim(M_{m,n}) = mn.$$

Parametrized subspace

For

$$W = \{(a + b, a - b, 2a) : a, b \in \mathbb{R}\},$$

we have

$$W = \text{span}\{(1, 1, 2), (1, -1, 0)\},$$

so $\dim(W) = 2$.

Polynomial subspace

A basis for $\{p \in P_3 : p(1) = 0\}$ is

$$\{x - 1, x^2 - x, x^3 - x^2\},$$

so its dimension is 3.

Closing note

Dimension counts independent directions; a basis is a minimal coordinate system for them.

5.1 Definitions and terminology

Term	Meaning
Basis	Independent spanning set.
Dimension	Number of vectors in any basis.
Basis extraction	Removal of redundant spanning vectors.

6 Rank of a Matrix and Systems of Linear Equations

Learning objectives

Find row and column space bases, rank, nullspace, nullity, and complete solution sets.

Let

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 \\ 2 & 1 & 3 & 1 & 3 \\ 1 & 0 & 2 & 2 & 5 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 3 \end{bmatrix}.$$

The nonzero rows of R form a row-space basis. Pivot columns 1, 2, 4 identify the original columns

$$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

as a column-space basis. Therefore $\text{rank}(A) = 3$.

Nullspace

The RREF equations give

$$\mathbf{x} = s \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \\ 0 \\ -3 \\ 1 \end{bmatrix}.$$

Thus nullity = 2, and $5 = 3 + 2$ verifies rank-nullity.

Nonhomogeneous solution

For $\mathbf{b} = (6, 12, 14)^T$, one particular solution is $\mathbf{x}_p = (1, 2, 0, -1, 3)^T$. Every solution is

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h,$$

where $\mathbf{x}_h$ is the nullspace expression above.

Column-space basis

Use pivot positions from the RREF, but choose the corresponding columns from the original matrix.

Closing note

Rank counts independent outputs; nullity counts input directions mapped to zero.

6.1 Definitions and terminology

Term	Meaning
Row space	Span of rows.
Column space	Span of columns.
Rank	Dimension of row or column space.

Nullspace	Solutions of $A\mathbf{x} = 0$.
Nullity	Dimension of the nullspace.
Rank-nullity	$n = \text{rank}(A) + \text{nullity}(A)$.

7 Coordinates and Change of Basis

Learning objectives

Find coordinate vectors, transition matrices, and coordinate representations in general spaces.

Coordinate vector

If $\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$ relative to ordered basis B , then

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}.$$

For $B = \{(1, 1), (1, -1)\}$ and $\mathbf{x} = (9, 21)$,

$$[\mathbf{x}]_B = \begin{bmatrix} 15 \\ -6 \end{bmatrix}.$$

Transition matrix

$P_{C \leftarrow B}$ satisfies $[\mathbf{x}]_C = P_{C \leftarrow B}[\mathbf{x}]_B$.

For

$$M_B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad M_C = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix},$$

$$P_{C \leftarrow B} = M_C^{-1}M_B = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}.$$

For $\mathbf{x} = (9, 20, 21)$,

$$[\mathbf{x}]_B = (4, 16, 5)^T, \quad [\mathbf{x}]_C = (-11, -1, 21)^T.$$

Polynomial coordinates

Relative to $\{1, x-1, (x-1)^2, (x-1)^3\}$,

$$2 + 3x - x^2 + x^3 = 5 + 4(x-1) + 2(x-1)^2 + (x-1)^3,$$

so the coordinate vector is $(5, 4, 2, 1)^T$.

Closing note

A basis change changes coordinate labels, not the underlying vector.

7.1 Definitions and terminology

Term	Meaning
Ordered basis	Basis with fixed vector order.
Coordinate vector	Coefficient list relative to an ordered basis.

Transition matrix

Converts coordinates between bases.

8 Applications of Vector Spaces

Learning objectives

Use Wronskians to test independence and rotate conics into principal-axis coordinates.

Wronskian

For functions $y_1, \dots, y_n$, the Wronskian is the determinant formed from the functions and derivatives through order $n - 1$.

For $y_1 = e^x$ and $y_2 = e^{-x}$,

$$W = \det \begin{bmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{bmatrix} = -2,$$

so the functions are independent solutions of $y'' - y = 0$. For $1, x, x^2$, the Wronskian is 2, giving the general solution of $y^{(3)} = 0$.

Standard conics

$$\frac{(x+1)^2}{9} + \frac{(y-2)^2}{4} = 1$$

is an ellipse centered at $(-1, 2)$ with semiaxes 3 and 2. The parabola $(x-2)^2 = -8(y-1)$ has vertex $(2, 1)$ and opens downward.

Rotation of axes

For

$$5x^2 - 6xy + 5y^2 = 20,$$

a 45° rotation yields

$$2x'^2 + 8y'^2 = 20,$$

or

$$\frac{x'^2}{10} + \frac{y'^2}{5/2} = 1.$$

Thus the original conic is a rotated ellipse.

Closing note

Wronskians convert functional independence to determinants; axis rotation aligns coordinates with a conic's natural geometry.

8.1 Definitions and terminology

Term	Meaning
Wronskian	Determinant of functions and derivatives.
Homogeneous differential equation	Linear equation with zero forcing term.
Conic section	Circle, ellipse, parabola, or hyperbola.
Rotation of axes	Orthogonal coordinate change eliminating an xy term.

Chapter summary

The chapter storyline

Vectors support linear combinations. Vector spaces axiomatize this behavior. Subspaces isolate linear subsets. Span and independence identify completeness and redundancy. Bases provide minimal coordinate systems, dimension counts them, rank and nullity describe matrix structure, change of basis converts coordinate descriptions, and Wronskians and conic rotations apply these ideas to functions and geometry.